

Santiago Hopkins

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EDUCATION

University of California, Berkeley

B.S. Mechanical Engineering, Student-Athlete (Soccer)

GPA: 3.82 / 4

EXPERIENCES

Triton Medical Robotics - *Robotics Hardware + Systems Engineering Intern*

01/2025 - Present

2-Axis Robot for Industrial Automation of Composite Tube Fabrication

- Led design of a fabrication robot (>200 parts) for 1.5 meter composite tube, decreasing fabrication time by 70%
- Selected industrial brushless motors for both spindle and linear axes, controlling for speed and torque via PID
- Conducted 5 time studies, studying time sinks to design hotfixes, allowing rapid prototyping of composite tube

Zoox - *Mechanical Design Engineering Intern*

05/2024 - 08/2024

Hydro-Pneumatic Sensor-Pod Wash System

- Led mechanical redesign of hydro pneumatic wash system, achieving ~98% cleaning efficiency
- Designed high production interfaces for external gear water pumps and normally-closed solenoid valves
- Integrated CAN communication for pump motor control, enabling 10hz pump motor state messaging
- Modeled hydraulic circuit in GT-Suite to influence future mechanical design, achieving ~88% accuracy

Water Contact Angle Measurement Software Automation Tool

- Developed software program to assess degradation of hydrophobic coatings on onboard camera lenses, reducing lab processing time by more than 60% and increasing measurement accuracy by 30%
- Implemented in Python, using OpenCV2 for computer vision and LSQ-Ellipse library for precise ellipse fitting
- Applied Gage R&R methods to validate on-fleet software automation tool, improving tool accuracy by ~20%

FSAE Driverless at Berkeley – *Co-Founder & Lead (Robotics - Mechanical)*

01/2023 -09/2024

- Selected and spearheaded a 4 member team in the 1st iteration of the mechanical systems that would steer and brake an FSAE Style EV
- ...becoming the **2nd American student team to have a fully-autonomous vehicle**

Autonomous Steering System ([docs](#))

- Led design and integration of robotic steering system (>100 parts), achieving lock to lock steering in <1s
- Specified system criteria, targeting 9Nm of motor torque, retaining low-profile design, minimizing backlash
- Designed belt drive with custom HTD-5mm pulleys and a rudimentary rotary encoder to prevent overturning
- Integrated servo motor via CAN and tuned PID controller, minimizing position error and overshoot

Autonomous Braking System ([docs](#))

- Led design of modular and redundant pneumatic braking system (>150 parts), achieving 100 brake actuations
- Designed Emergency Brake System via clever mechanical & electrical design, achieving 180ms response time
- Developed cold-swappable pneumatic circuit, allowing easy switching between manual and autonomous brakes; via shuttle valve, simultaneous manual and autonomous braking is retained for test driving purposes

World Surf League – *Systems Engineering Intern*

05/2023 -12/2023

- Computationally and experimentally determined effect of on-pole LED strip lights on overall waveshape, saved several thousands of dollars and prevented week long delay by avoiding light removal
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PROJECTS

DIY Autonomous Drone

07/2024 - 12/2024

- Led mechanical design of DIY drone, achieving controlled motion over a 1m distance at a height of 25cm
 - Implemented global path planning and 'carrot-on-a-stick' method for motion planning
 - Developed 3D printed fixtures to protect onboard compute, maintaining symmetry and minimizing weight
 - Designed test fixtures, constraining drone to streamline Nonlinear PID tuning; programmed crucial fail safes
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SKILLS

Hardware: Onshape, SolidWorks, CATIA v5/6, Ansys (structural, thermal, vibration), PLM, AutoCad, DFM/A

Software: Python, C++, ROS2, Linux, Git, Robotics (Kinematics, Controls, Sims), MATLAB, Node JS, C, Java

Manufacturing: GD&T, CNC Machining, Injection Molding, Casting, Waterjet, Laser Cutter, FDM/SLA, 3D Printing, Lathe, Mill, Welding, Sheet Metal